

My Report

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Table of contents

0.1 Setup and Cleaning	1
1 Setup	1
2 Frequentist model confidence intervals and significance	1
3 Bayesian Model with priors	2
3.1 Post summary with credible intervals	2
4 Credible vs Confidence intervals	3
5 Interpretations	4
6 Log NIL from BPR and conference	5

0.1 Setup and Cleaning

1 Setup

2 Frequentist model confidence intervals and significance

term	estimate	lower	upper	p_value	pct_estimate	pct_lower	pct_upper	s
str	f64	f64	f64	f64	f64	f64	f64	b
"conference_ACC"	0.164	-0.161	0.488	0.321	17.783	-14.843	62.908	f
"conference_BigEast"	0.158	-0.213	0.529	0.402	17.137	-19.15	69.71	f
"conference_SEC"	0.098	-0.222	0.419	0.546	10.347	-19.929	52.071	f
"conference_Big12"	0.053	-0.276	0.383	0.75	5.485	-24.104	46.611	f
"bpr"	0.028	0.018	0.039	0.0	2.867	1.804	3.941	t
"conference_Atlantic10"	-0.091	-0.443	0.262	0.613	-8.675	-35.815	29.94	f
"conference_WestCoast"	-0.403	-0.798	-0.009	0.045	-33.2	-54.969	-0.907	t
"conference_Pac12"	-0.468	-0.845	-0.09	0.015	-37.358	-57.045	-8.647	t

term str	estimate f64	lower f64	upper f64	p_value f64	pct_estimate f64	pct_lower f64	pct_upper f64	s
"conference_American"	-0.555	-0.974	-0.136	0.01	-42.594	-62.247	-12.71	t
"conference_MEAC"	-0.677	-1.268	-0.086	0.025	-49.189	-71.863	-8.243	t
"conference_SunBelt"	-0.812	-1.226	-0.398	0.0	-55.604	-70.654	-32.837	t
"conference_MountainWest"	-0.908	-1.339	-0.478	0.0	-59.685	-73.789	-37.99	t
"conference_MAC"	-1.025	-1.412	-0.638	0.0	-64.12	-75.629	-47.175	t
"conference_ConferenceUSA"	-1.044	-1.444	-0.644	0.0	-64.799	-76.404	-47.485	t
"conference_BigWest"	-1.18	-1.582	-0.778	0.0	-69.276	-79.447	-54.069	t
"conference_MissouriValley"	-1.233	-1.648	-0.817	0.0	-70.853	-80.762	-55.841	t
"conference_SoCon"	-1.344	-1.83	-0.858	0.0	-73.914	-83.952	-57.596	t
"conference_SummitLeague"	-1.346	-1.81	-0.882	0.0	-73.97	-83.631	-58.608	t
"conference_CAA"	-1.403	-1.812	-0.995	0.0	-75.42	-83.66	-63.025	t
"conference_MAAC"	-1.474	-1.887	-1.06	0.0	-77.092	-84.847	-65.366	t
"conference_Horizon"	-1.484	-1.911	-1.057	0.0	-77.333	-85.208	-65.264	t
"conference_ASUN"	-1.601	-2.026	-1.177	0.0	-79.833	-86.81	-69.165	t
"conference_BigSouth"	-1.636	-2.081	-1.191	0.0	-80.519	-87.517	-69.597	t
"conference_SWAC"	-1.661	-2.132	-1.189	0.0	-80.998	-88.14	-69.556	t
"conference_BigSky"	-1.722	-2.163	-1.28	0.0	-82.122	-88.506	-72.192	t
"conference_NEC"	-1.806	-2.332	-1.279	0.0	-83.563	-90.292	-72.172	t
"conference_OhioValley"	-1.833	-2.338	-1.328	0.0	-84.011	-90.351	-73.504	t
"conference_AmericaEast"	-1.85	-2.303	-1.398	0.0	-84.282	-90.004	-75.284	t
"conference_Patriot"	-1.858	-2.442	-1.274	0.0	-84.398	-91.302	-72.017	t
"conference_WAC"	-1.858	-2.311	-1.405	0.0	-84.404	-90.088	-75.461	t
"conference_Southland"	-2.09	-2.514	-1.666	0.0	-87.633	-91.907	-81.104	t
"conference_IvyLeague"	-2.264	-2.701	-1.828	0.0	-89.61	-93.286	-83.921	t

3 Bayesian Model with priors

```

Initializing NUTS using jitter+adapt_diag...
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/pytensor/tensor
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, bpr, conference_ACC, conference_ASUN, conference_AmericaEast, conferen

```

Output()

Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws total) took 3 s

3.1 Post summary with credible intervals

	mean	sd	hdi95_lb	hdi95_ub	ess_bulk	ess_tail	r_hat	mcse_r
sigma	0.4352	0.0197	0.3968	0.4744	8144.0408	5364.5352	1.0000	0.0002
Intercept	1.4122	0.1415	1.1392	1.7005	561.6913	833.6019	1.0113	0.0059
bpr	0.0284	0.0052	0.0184	0.0387	1345.5196	2619.2812	1.0027	0.0001
conference_ACC	0.1675	0.1653	-0.1530	0.4968	1257.3440	2299.5886	1.0061	0.0047
conference_ASUN	-1.5941	0.2126	-2.0070	-1.1680	964.9353	1722.6803	1.0058	0.0068
conference_AmericaEast	-1.8418	0.2270	-2.2920	-1.4051	980.2853	1777.6176	1.0056	0.0072
conference_American	-0.5489	0.2124	-0.9616	-0.1301	1386.2572	2060.4060	1.0040	0.0057
conference_Atlantic10	-0.0836	0.1789	-0.4299	0.2741	1087.5047	1727.8221	1.0059	0.0054
conference_Big12	0.0595	0.1682	-0.2722	0.3794	1638.0002	3045.1731	1.0049	0.0042
conference_BigEast	0.1633	0.1886	-0.1897	0.5367	1670.6580	3082.3414	1.0047	0.0046
conference_BigSky	-1.7153	0.2259	-2.1713	-1.2817	1046.6995	2025.0920	1.0063	0.0070
conference_BigSouth	-1.6278	0.2256	-2.0814	-1.1945	1179.7329	2151.3452	1.0045	0.0066
conference_BigWest	-1.1732	0.2039	-1.5591	-0.7674	919.4155	1904.7805	1.0063	0.0067
conference_CAA	-1.3964	0.2050	-1.7998	-0.9987	966.0994	1745.7157	1.0063	0.0066
conference_ConferenceUSA	-1.0369	0.2000	-1.4289	-0.6554	976.3161	1550.6399	1.0063	0.0064
conference_Horizon	-1.4757	0.2124	-1.8754	-1.0370	956.3084	1799.0213	1.0056	0.0069
conference_IvyLeague	-2.2570	0.2168	-2.6900	-1.8425	1255.8856	2036.0413	1.0049	0.0061
conference_MAAC	-1.4654	0.2082	-1.8852	-1.0760	915.0133	1545.1839	1.0064	0.0069
conference_MAC	-1.0181	0.1948	-1.4008	-0.6404	995.4084	1718.7370	1.0065	0.0062
conference_MEAC	-0.6701	0.2975	-1.2651	-0.0989	1262.1770	2244.3104	1.0040	0.0084
conference_MissouriValley	-1.2262	0.2068	-1.6381	-0.8393	1173.3141	2102.7489	1.0052	0.0060
conference_MountainWest	-0.9010	0.2141	-1.3220	-0.4851	1590.7698	2724.9625	1.0033	0.0054
conference_NEC	-1.8006	0.2668	-2.3475	-1.2958	1196.7371	2150.0535	1.0047	0.0077
conference_OhioValley	-1.8261	0.2553	-2.3254	-1.3258	1237.7501	2007.2542	1.0037	0.0072
conference_Pac12	-0.4610	0.1913	-0.8427	-0.0975	1463.8393	2568.0591	1.0040	0.0050
conference_Patriot	-1.8504	0.2982	-2.4486	-1.2842	1769.1730	3122.4411	1.0039	0.0071
conference_SEC	0.1033	0.1625	-0.2111	0.4221	1549.5931	2957.8004	1.0049	0.0041
conference_SWAC	-1.6536	0.2374	-2.1273	-1.1948	951.1729	1388.1973	1.0062	0.0077
conference_SoCon	-1.3370	0.2435	-1.8153	-0.8515	1255.0438	2600.8155	1.0058	0.0069
conference_Southland	-2.0838	0.2134	-2.5171	-1.6843	1044.2827	1753.4820	1.0058	0.0066
conference_SummitLeague	-1.3376	0.2318	-1.7698	-0.8669	1264.2476	2597.5361	1.0039	0.0065
conference_SunBelt	-0.8046	0.2118	-1.2158	-0.3860	992.0464	1926.8277	1.0064	0.0067
conference_WAC	-1.8505	0.2265	-2.2729	-1.3752	1126.6176	2007.7933	1.0041	0.0067
conference_WestCoast	-0.3968	0.1998	-0.7949	-0.0048	1060.5887	2002.4399	1.0059	0.0061

With many dummy variables the default settings gave low effective sample sizes and a r-hat warning, so we raised the draws a bunch and tuning steps. In the summary, every r_hat is at most 1.004, the smallest ess_bulk is about 480, and there were no divergences. That means the chains converged.

4 Credible vs Confidence intervals

term str	conf_lower f64	conf_upper f64	cred_lower f64	cred_upper f64	conf_significant bool	cred_s bool
"Intercept"	1.137	1.7	1.139	1.7	true	true
"bpr"	0.018	0.039	0.018	0.039	true	true
"conference_ACC"	-0.161	0.488	-0.153	0.497	false	false
"conference_ASUN"	-2.026	-1.177	-2.007	-1.168	true	true
"conference_AmericaEast"	-2.303	-1.398	-2.292	-1.405	true	true
"conference_American"	-0.974	-0.136	-0.962	-0.13	true	true
"conference_Atlantic10"	-0.443	0.262	-0.43	0.274	false	false
"conference_Big12"	-0.276	0.383	-0.272	0.379	false	false
"conference_BigEast"	-0.213	0.529	-0.19	0.537	false	false
"conference_BigSky"	-2.163	-1.28	-2.171	-1.282	true	true
"conference_BigSouth"	-2.081	-1.191	-2.081	-1.194	true	true
"conference_BigWest"	-1.582	-0.778	-1.559	-0.767	true	true
"conference_CAA"	-1.812	-0.995	-1.8	-0.999	true	true
"conference_ConferenceUSA"	-1.444	-0.644	-1.429	-0.655	true	true
"conference_Horizon"	-1.911	-1.057	-1.875	-1.037	true	true
"conference_IvyLeague"	-2.701	-1.828	-2.69	-1.842	true	true
"conference_MAAC"	-1.887	-1.06	-1.885	-1.076	true	true
"conference_MAC"	-1.412	-0.638	-1.401	-0.64	true	true
"conference_MEAC"	-1.268	-0.086	-1.265	-0.099	true	true
"conference_MissouriValley"	-1.648	-0.817	-1.638	-0.839	true	true
"conference_MountainWest"	-1.339	-0.478	-1.322	-0.485	true	true
"conference_NEC"	-2.332	-1.279	-2.348	-1.296	true	true
"conference_OhioValley"	-2.338	-1.328	-2.325	-1.326	true	true
"conference_Pac12"	-0.845	-0.09	-0.843	-0.098	true	true
"conference_Patriot"	-2.442	-1.274	-2.449	-1.284	true	true
"conference_SEC"	-0.222	0.419	-0.211	0.422	false	false
"conference_SWAC"	-2.132	-1.189	-2.127	-1.195	true	true
"conference_SoCon"	-1.83	-0.858	-1.815	-0.852	true	true
"conference_Southland"	-2.514	-1.666	-2.517	-1.684	true	true
"conference_SummitLeague"	-1.81	-0.882	-1.77	-0.867	true	true
"conference_SunBelt"	-1.226	-0.398	-1.216	-0.386	true	true
"conference_WAC"	-2.311	-1.405	-2.273	-1.375	true	true
"conference_WestCoast"	-0.798	-0.009	-0.795	-0.005	true	true

same_significance bool	max_gap_lower f64	max_gap_upper f64	mean_width_ratio f64
true	0.04	0.03	0.986003

5 Interpretations

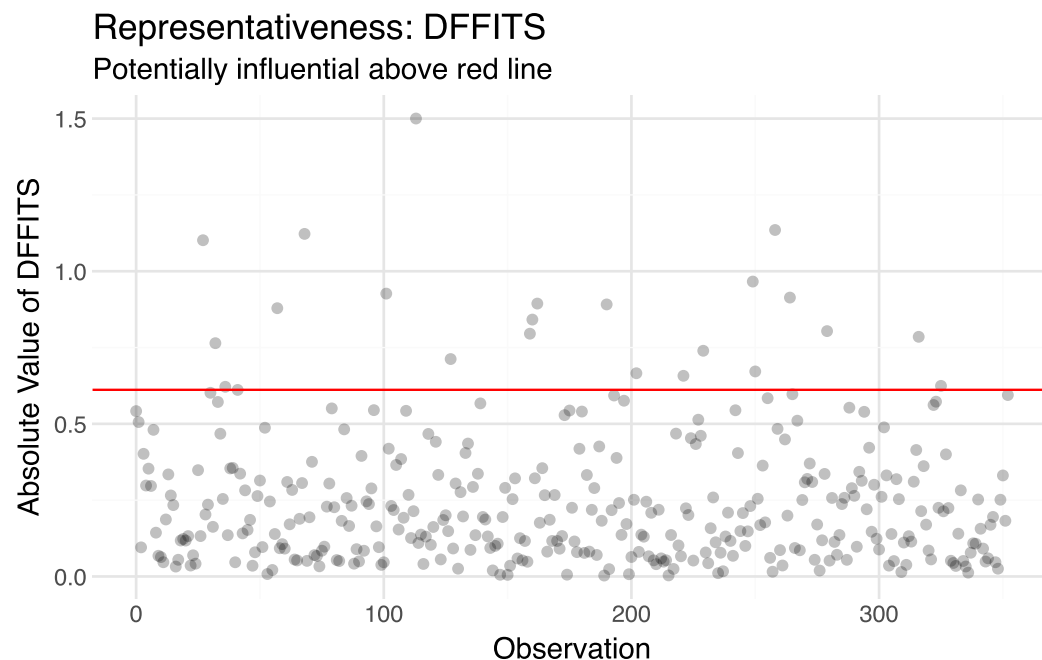
- BPR: The slope for the confidence interval is 0.018, 0.039 on the log scale, with $p < 0.001$

- Confidence Interval: Holding all other variables fixed, Based on the OLS interval estimate, 95% of such intervals will contain the true value, so we are 95% confident that one more BPR point increases team NIL by between 1.8% and 3.9% (point estimate 2.9%), holding conference fixed.
- Credible Interval: Holding all other variables fixed, The 95% credible interval is (0.0185, 0.0386). With probability 0.95, one more BPR point increases team NIL by between 1.9% and 3.9%
- Significance: The interval does not include 0, so BPR is statistically significant
- Intercept

Holding all other variables fixed, A Big Ten team with a BPR of 0 has an expected NIL of \$4.13M (95% CI \$3.12M to \$5.47M). This is a baseline only, since no Big Ten team has a BPR near 0.

- Conferences Holding all other variables fixed, 27 of the 32 slope estimates are statistically significant, BPR and 26 conferences among them.
- Credible vs Confidence The two intervals are almost exactly the same. The largest endpoint gap on the log scale is 0.035, the Bayesian model intervals are on avg 99% as wide as the OLS ones. And they both agree on significance for all 33 of the terms. The remaining difference comes from sampling noise and from HDI vs the OLS interval. Interpretation is different though. The credible interval is a direct probability statement about the parameter, given the data and model. The confidence interval describes how often the procedure would capture the true value across repeated samples.

6 Log NIL from BPR and conference



team	tier	bpr	nil_m	dffits
str	str	f64	f64	f64
"Drake"	"Basketball"	3.68482	4.45	1.12231
"Bradley"	"Basketball"	-2.1846	3.8	1.101547
"Saint Mary's"	"LowerTier"	4.977795	9.35	0.96613
"Gonzaga"	"G5"	25.032172	12.1	0.926665
"South Carolina Upstate"	"FCS"	-14.566598	1.55	0.91364
"Memphis"	"G5"	5.775717	9.2	0.894015
"New Orleans"	"LowerTier"	-8.632951	1.15	0.891231
"Coppin State"	"FCS"	-15.098677	2.25	0.87887
"Massachusetts"	"G5"	0.062609	4.85	0.841557
"Stetson"	"LowerTier"	-8.66753	1.75	0.803797
"Maryland-Eastern Shore"	"FCS"	-18.023558	1.9	0.795364
"UNLV"	"G5"	5.745695	3.9	0.785122
"Jacksonville"	"LowerTier"	-6.714238	1.65	0.712268
"Saint Peter's"	"Basketball"	-8.656443	1.9	0.67164
"Northeastern"	"FCS"	-5.288444	2.0	0.665763
"Cal State Northridge"	"LowerTier"	-7.180107	2.15	0.621364
"VCU"	"Basketball"	14.128465	2.4	-0.624408
"Pacific"	"LowerTier"	-0.32085	1.3	-0.657391
"Presbyterian"	"FCS"	-2.788702	0.3	-0.739391
"Butler"	"Basketball"	6.176608	2.35	-0.764263
"Seattle"	"LowerTier"	1.601404	0.8	-1.13507
"Howard"	"FCS"	-1.411592	0.3	-1.500268